

PALMER, AK, USA

WMO: 702740

Lat: 61.596N

Lon: 149.092W

Elev: 70

StdP: 100.49

Time Zone: -9.00 (NAK)

Period: 97-19

WBAN: 25331

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-22.5	-28.9	0.3	-23.5	-26.3	0.3	-19.4	18.4	-5.6	16.4	-3.7	0.7	0	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	23.9	15.8	22.1	14.7	20.2	13.8	16.6	23.0	15.4	20.9	14.5	19.1	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.5	9.7	18.7	12.6	9.2	17.6	12.0	8.8	16.8	46.3	23.1	43.2	20.9	40.5	19.2	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	11.2	9.0	-27.0	26.1	4.4	2.2	-30.1	27.6	-32.7	28.9	-35.1	30.2	-38.3	31.8		
			-27.1	17.9	4.2	1.2	-30.1	18.8	-32.6	19.5	-35.0	20.1	-38.0	21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-8.6	-5.9	-3.2	3.6	9.4	13.4	15.0	13.5	9.2	2.5	-5.2	-7.5
	DBStd	9.89	8.13	6.66	5.49	3.72	3.00	2.33	2.04	2.17	2.75	4.77	6.57	7.04
	HDD10.0	2947	576	445	408	192	46	1	0	1	44	234	456	544
	HDD18.3	5572	834	678	666	441	276	149	104	151	273	491	706	802
	CDD10.0	419	0	0	0	1	29	103	156	109	21	2	0	0
	CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0
	CDH23.3	73	0	0	0	0	3	21	36	13	0	0	0	0
	CDH26.7	6	0	0	0	0	0	2	3	1	0	0	0	0
Wind	WSAvg	3.0	3.0	3.1	3.4	3.7	3.6	2.9	2.5	2.4	2.7	2.8	2.9	3.1
Precipitation	PrecAvg	414	25	23	16	11	17	29	50	66	75	38	29	35
	PrecMax	551	66	69	54	35	62	65	107	157	147	76	64	92
	PrecMin	302	2	3	0	0	0	0	10	21	25	12	3	1
	PrecStd	67	15	16	12	8	14	17	25	32	37	19	18	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	8.0	9.4	17.1	23.4	26.0	26.5	25.0	18.2	14.9	9.1	7.8
		MCWB	4.2	4.1	5.0	9.3	13.0	16.3	17.0	16.7	11.9	9.4	5.4	4.1
	2%	DB	6.1	5.5	7.1	13.4	20.4	23.0	24.1	22.5	16.4	11.8	7.2	6.0
		MCWB	3.1	2.4	3.0	6.5	11.3	14.9	16.4	15.3	11.2	7.4	4.2	3.2
	5%	DB	4.1	4.0	5.5	11.4	17.9	21.1	22.1	20.1	15.1	10.0	5.1	3.8
		MCWB	1.7	1.6	2.0	5.3	10.3	14.1	15.4	14.4	10.5	6.4	2.7	1.5
	10%	DB	2.2	2.6	4.0	9.8	15.9	19.0	20.2	18.2	13.8	8.7	3.2	1.9
		MCWB	0.1	0.4	1.0	4.6	9.1	12.8	14.4	13.2	9.9	5.6	1.2	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.5	5.2	9.6	13.5	17.2	18.0	17.5	12.9	9.8	5.9	4.5
		MCDB	7.7	7.6	9.0	16.7	22.3	24.3	25.0	23.6	16.6	14.2	8.3	7.1
	2%	WB	3.3	2.9	3.6	7.1	11.8	15.7	16.8	16.0	11.9	8.1	4.3	3.3
		MCDB	5.8	5.0	6.3	12.1	19.2	22.0	23.4	21.5	15.3	10.9	6.9	5.8
	5%	WB	1.9	1.7	2.4	5.8	10.6	14.3	15.7	14.9	11.1	6.9	3.0	1.7
		MCDB	4.1	3.8	5.0	10.6	17.0	20.0	21.4	19.3	14.1	9.4	4.9	3.5
	10%	WB	0.1	0.5	1.3	4.9	9.5	13.2	14.8	13.8	10.4	5.9	1.4	-0.1
		MCDB	2.0	2.5	3.8	9.3	15.3	18.3	19.6	17.5	13.1	8.3	3.1	1.6
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.1	8.1	9.1	10.3	9.4	8.7	8.3	7.7	6.7	6.2	6.4
		MCDBR	6.7	6.9	7.9	12.0	14.3	13.5	13.0	12.1	9.4	7.3	6.4	7.7
		MCWBR	4.9	5.3	5.5	6.9	7.3	7.0	6.8	6.7	5.4	4.5	4.7	6.0
	5% WB	MCDBR	6.5	6.4	7.2	11.1	13.1	12.4	11.8	10.7	8.2	6.5	6.1	7.1
		MCWBR	5.0	5.2	5.2	6.6	6.8	6.8	6.5	6.3	5.1	4.3	4.6	5.9
Clear-Sky Solar Irradiance	taub	0.229	0.286	0.302	0.342	0.364	0.370	0.374	0.359	0.313	0.288	0.241	N/A	
	taud	2.071	2.151	2.194	2.152	2.195	2.312	2.344	2.396	2.493	2.336	2.035	N/A	
	Ebn at Noon	481	673	808	841	851	854	840	823	808	680	433	N/A	
	Edh at Noon	37	69	94	121	126	115	109	95	71	57	37	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.98	2.61	4.38	5.21	5.33	4.75	3.77	2.45	1.13	0.41	0.14	
	RadStd	0.04	0.11	0.24	0.34	0.40	0.34	0.36	0.38	0.31	0.12	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	+1.98	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page